STPM 2015 PHYSICS 960/2 (U2)

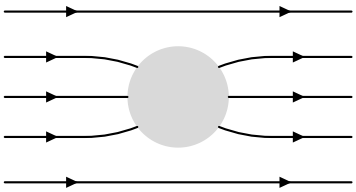
Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $g = 9.81$ m s^{-2}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

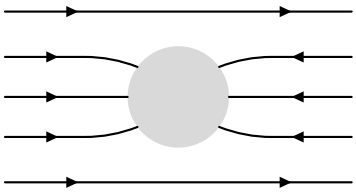
Section A [15 marks]

1. Which diagram shows the correct electric field lines when a spherical conductor is placed in a uniform electric field?

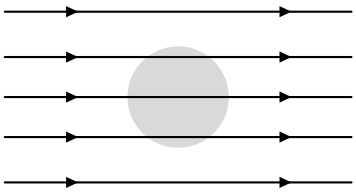
A



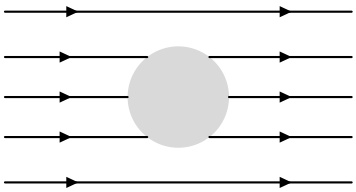
B



C



D



Answer: A

Field lines end on (and start from) the conductor's surface at right angles, and there is no field inside the conductor (A).

Explanation: Induced charges on the conductor cancel the field inside it.

2. Spherical conductor *Y* is larger than spherical conductor *X*. Conductor *X* is initially uncharged whereas conductor *Y* is charged. A wire is connected from conductor *Y* to *X*. Which statement is true when equilibrium is attained?

A Conductors *X* and *Y* have the same electric potentials.
B Conductors *X* and *Y* have the same quantity of charges.
C Conductors *X* and *Y* have the same surface charge densities.
D Conductors *X* and *Y* have the same electric field strengths at the surface.

Answer: A

Charge flows through the wire until there is no p.d. between the conductors: they reach the same electric potential (A).

Explanation: The larger sphere ends up with more charge but a smaller surface charge density.

3. A parallel plate capacitor is fully charged. When an insulator is inserted into the space between the plates of the capacitor, the energy stored in the capacitor decreases by 73.0%. What is the value of the dielectric constant of the insulator?

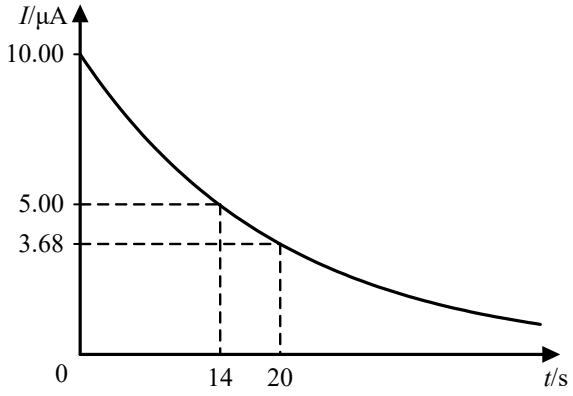
A 1.37 B 1.88 C 3.70 D 13.7

Answer: C

Charge constant: $U = \frac{Q^2}{2C} \propto \frac{1}{C}$; $U' = 0.270U \Longrightarrow \varepsilon_r = \frac{1}{0.270} = 3.70$

Explanation: The capacitor is isolated after charging, so Q does not change.

4. The variation of current I with time t when a capacitor is discharging through a resistor in a circuit is shown in the graph below.



What is the time constant of the circuit?

A 6.0 s B 14 s C 20 s D 34 s

Answer: C

After one time constant $I = 0.368I_0 = 3.68$ μA, which the graph shows at $t = 20$ s

Explanation: At 14 s the current has halved: 14 s = τ ln 2.

5. Which statement about conduction of electricity is **not** true?

A Metals are generally better electrical conductors because they have more free electrons.
B Conductivity of metals increases at a higher temperature because the free electrons move faster.
C Copper is a better conductor than aluminium because copper has a higher charge carrier density.
D Conductivity of semiconductor increases at a higher temperature because the number of charge carriers increases.

Answer: B

The conductivity of a metal decreases at higher temperature (more collisions), so statement B is not true.

Explanation: Only in semiconductors does a rise in temperature increase conductivity.

6. Potential difference V is supplied across a uniform wire of length L and cross-sectional area A. The number of free electrons per unit volume is n and the resistivity of the wire is ρ. If e is the charge of an electron, the drift velocity of free electrons is

A $\frac{V}{\rho Lne}$ B $\frac{1}{\rho LVne}$ C $\frac{VL}{\rho nA^2e}$ D $\frac{\rho L}{VnA^2e}$

Answer: A

$$J = \frac{E}{\rho} = \frac{V}{\rho L} = nev \Longrightarrow v = \frac{V}{\rho Lne}$$

Explanation: The area cancels.

7. A resistor R is connected to a cell of e.m.f. ε and internal resistance r. The terminal potential difference of the cell is V. The expression for the internal resistance r of the cell is

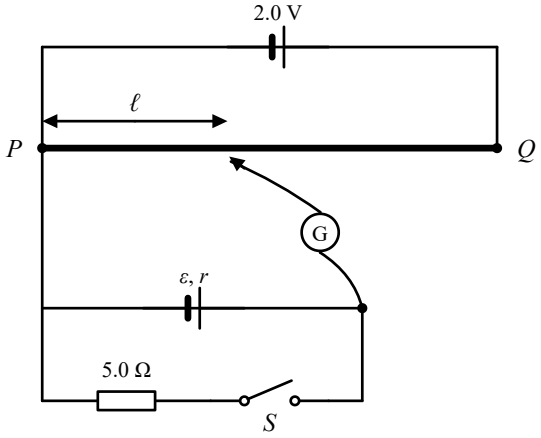
A $r = R(\frac{\varepsilon}{V})$ B $r = R(\frac{\varepsilon}{V} - 1)$ C $r = R(\frac{V}{\varepsilon} - 1)$ D $r = R(\frac{V}{\varepsilon - V})$

Answer: B

$$\varepsilon = V + Ir \text{ with } I = \frac{V}{R} \Longrightarrow r = \frac{R(\varepsilon - V)}{V} = R(\frac{\varepsilon}{V} - 1)$$

Explanation: The lost volts $\varepsilon - V$ appear across r.

8. A potentiometer consisting of a uniform slide wire PQ of length 100 cm is connected to a power supply of 2.0 V as shown in the diagram below. The internal resistance of the driver cell is negligible.



When switch S is opened, the balanced length ℓ is 80.0 cm and when the switch is closed, the balanced length is 50.0 cm. What are the values of ε and r?

	ε	r
A	1.0 V	1.3 Ω
B	1.0 V	3.0 Ω
C	1.6 V	1.3 Ω
D	1.6 V	3.0 Ω

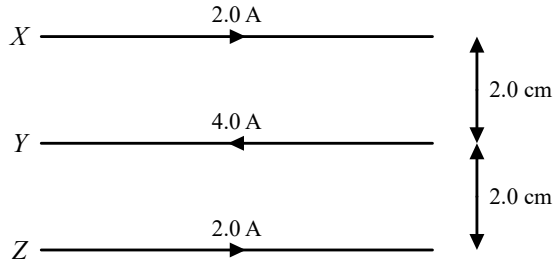
Answer: D

$$\text{Gradient} = \frac{2.0}{100} = 0.020 \text{ V cm}^{-1}; \; S \text{ open: } \varepsilon = (0.020)(80.0) = 1.6 \text{ V}$$

$$S \text{ closed: } V = (0.020)(50.0) = 1.0 \text{ V}; \; r = 5.0\left(\frac{1.6}{1.0} - 1\right) = 3.0 \; \Omega$$

Explanation: With *S* open no current flows in the cell at balance.

9. Three long conductors, *X*, *Y* and *Z*, are placed parallel to each other at a distance of 2.0 cm as shown in the diagram below.



Currents flow in *X*, *Y* and *Z* are 2.0 A, 4.0 A and 2.0 A respectively. What are the magnitude and the direction of the resultant force per unit length acting on *X*?

	Magnitude	*Direction*
A	10×10^{-5} N m^{-1}	Upward
B	10×10^{-5} N m^{-1}	Downward
C	6.0×10^{-5} N m^{-1}	Upward
D	6.0×10^{-5} N m^{-1}	Downward

Answer: C

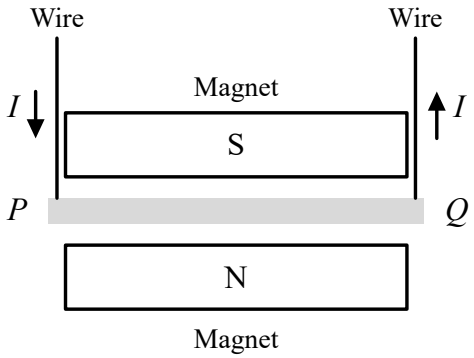
$$\text{From } Y \text{ (opposite current, repulsion, upward): } \frac{\mu_0(2.0)(4.0)}{2\pi(0.020)} = 8.0 \times 10^{-5} \text{ N m}^{-1}$$

$$\text{From } Z \text{ (same direction, attraction, downward): } \frac{\mu_0(2.0)(2.0)}{2\pi(0.040)} = 2.0 \times 10^{-5} \text{ N m}^{-1}$$

Resultant = 6.0×10^{-5} N m^{-1} upward.

Explanation: *Z* is 4.0 cm from *X*.

10. A copper rod PQ of density 8900 kg m^{-3} and cross-sectional area 1.3×10^{-7} m^2 is suspended horizontally between two permanent magnets with magnetic flux density 7.9×10^{-2} T as shown in the diagram below.



What is the current I flowing in the rod so that the tension of the supporting wires is reduced to zero?

A 0.015 A B 0.14 A C 0.57 A D 7.1 A

Answer: B

$$BIL = mg = \rho ALg \Longrightarrow I = \frac{\rho Ag}{B} = \frac{(8900)(1.3 \times 10^{-7})(9.81)}{7.9 \times 10^{-2}} = 0.14 \text{ A}$$

Explanation: The length of the rod cancels.

11. Which statement is true when a charge moves in a uniform magnetic field?

A It will always experience a magnetic force.
B It will always move with a constant velocity.
C It will gain kinetic energy due to the work done by magnetic force.
D Its path is circular when it moves perpendicularly to the direction of the field.

Answer: D

A charge moving perpendicular to a uniform magnetic field moves in a circle (D).

Explanation: A charge moving along the field feels no force; the magnetic force never changes the kinetic energy.

12. An inductor consists of 20 000 turns and radius 3.7 cm. If the length of the inductor is 0.45 m, the self inductance is

A 1.2 H B 2.2 H C 4.8 H D 11 H

Answer: C

$$L = \frac{\mu_0 N^2 A}{l} = \frac{(4\pi \times 10^{-7})(20\,000)^2 \pi (0.037)^2}{0.45} = 4.8 \text{ H}$$

Explanation: Square the number of turns.

13. A coil has a self inductance of 3.0 mH. If the current in the coil is reduced from 25 A to zero in 0.50 s, the induced e.m.f. in the coil is

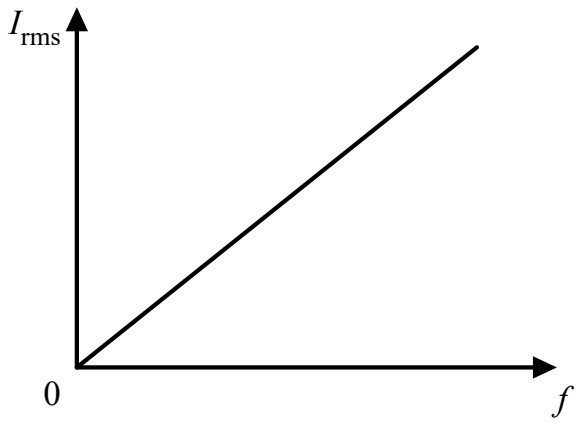
A 17 mV B 38 mV C 75 mV D 150 mV

Answer: D

$$\varepsilon = L\frac{\Delta I}{\Delta t} = (3.0 \times 10^{-3})\frac{25}{0.50} = 0.15 \text{ V} = 150 \text{ mV}$$

Explanation: The rate of change of current is 50 A s^{-1}.

14. A sinusoidal voltage supply with a constant r.m.s. voltage but variable frequency f is connected in series with an unknown electrical component. The variation of r.m.s. current I_{rms} with the frequency f is shown in the graph below.



The electrical component is

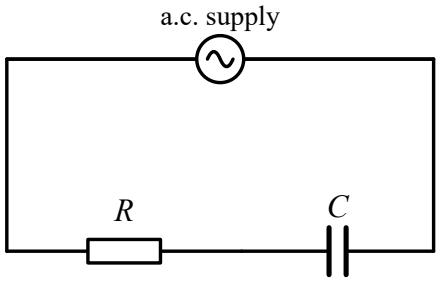
A a resistor B a capacitor C an inductor D a diode

Answer: B

$$I_{rms} = \frac{V_{rms}}{X_C} = V_{rms}(2\pi fC) \propto f\text{: a capacitor}$$

Explanation: For an inductor the current would fall as $1/f$; for a resistor it would be constant.

15. A capacitor of capacitance C and a resistor of resistance R are connected in series with an a.c. supply of frequency f as shown in the diagram below.



Which expression represents the impedance Z of the circuit?

A $R + 2\pi fC$ B $R + \frac{1}{2\pi fC}$ C $\sqrt{(R^2 + (2\pi fC)^2)}$ D $\sqrt{(R^2 + (\frac{1}{2\pi fC})^2)}$

Answer: D

$$Z = \sqrt{(R^2 + X_C^2)} \text{ with } X_C = \frac{1}{2\pi fC} \text{ (D)}$$

Explanation: R and X_C are 90° out of phase, so they add as phasors.

Section B [15 marks]

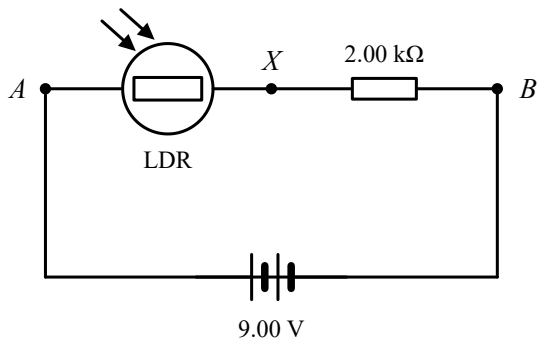
16. (a) State the function and the operation of a potential divider in an electric circuit. [2]

Answer

Function: to supply a chosen fraction of the supply voltage to a load	1
Operation: resistors in series share the supply p.d. in the ratio of their resistances	1
	2

Explanation: $V_{out} = V_{in}R_2/(R_1 + R_2)$.

(b) A light dependent resistor (LDR) has a minimum resistance of 1.00 kΩ in bright light as shown in the diagram below.



The LDR is connected in series to a resistor of 2.00 kΩ and a voltage supply of 9.00 V.

(i) Determine the voltage V_{AX} across the LDR in bright light. [2]

Answer

$V_{AX} = 9.00 \times \frac{1.00}{1.00 + 2.00}$ 1

$V_{AX} = 3.00$ V 1

2

Explanation: The p.d. divides in the ratio 1 : 2.

(ii) Calculate the resistance of R if V_{AX} is 2.75 V in bright light. [1]

Answer

$\frac{2.75}{1.00} = \frac{9.00 - 2.75}{R} \Rightarrow R = 2.27$ kΩ 1

1

Explanation: The current is the same through both resistors.

(iii) If another similar LDR is placed in parallel to the first LDR, calculate V_{AX} in bright light. [2]

Answer

Two LDRs in parallel: 0.500 kΩ 1

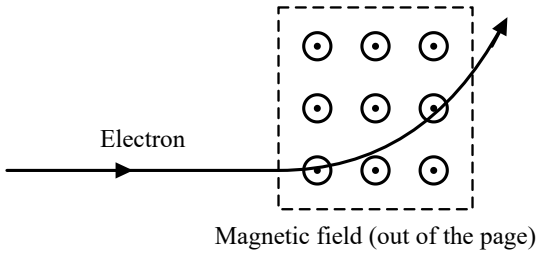
$V_{AX} = 9.00 \times \frac{0.500}{0.500 + 2.00} = 1.80$ V 1

2

Total for question **7**

Explanation: A smaller resistance takes a smaller share of the p.d.

17. The path of an electron moving through a uniform magnetic field perpendicularly out of the plane of the paper is shown in the diagram below.



The speed of the electron is 4.8×10^6 m s^{-1} and the magnetic flux density is 5.6×10^{-3} T. The radius of the path of the electron in the magnetic field is 4.88×10^{-3} m.

(i) Determine the ratio $\frac{e}{m}$ for the electron. [4]

Answer

$evB = \frac{mv^2}{r}$ 1

$\frac{e}{m} = \frac{v}{Br}$ 1

$\frac{e}{m} = \frac{4.8 \times 10^6}{(5.6 \times 10^{-3})(4.88 \times 10^{-3})}$ 1

$= 1.76 \times 10^{11}$ C kg^{-1} 1

4

Explanation: The magnetic force provides the centripetal force.

(ii) Determine the magnitude and direction of an electric field which must be applied on the electron so that the direction of its movement does not change. [4]

Answer

For no deflection the electric force balances the magnetic force: $eE = evB$ 1

$E = vB = (4.8 \times 10^6)(5.6 \times 10^{-3})$ 1

$E = 2.69 \times 10^4$ V m^{-1} 1

The magnetic force on the electron is upwards, so the electric force must be downwards: **E** points upwards (towards the top of the page) 1

4

Total for question **8**

Explanation: The force on an electron is opposite to **E**.

Section C [30 marks]

18. (a) State Coulomb's law. [1]

Answer

The force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of their separation: $F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$ 1

1

Explanation: The force acts along the line joining the charges.

(b) Two particles of charge 2.05 μC and 4.50 μC are placed 5.00 cm apart in air. Determine

(i) the electric force experienced by each of the particles, [2]

Answer

$F = \dfrac{(9.0 \times 10^9)(2.05 \times 10^{-6})(4.50 \times 10^{-6})}{(0.0500)^2}$ 1

$F = 33.2$ N, repulsive 1

2

Explanation: Both charges are positive.

(ii) the electric field at the midpoint between the particles, [2]

Answer

$E = \dfrac{k(4.50 - 2.05) \times 10^{-6}}{(0.0250)^2}$ 1

$E = 3.53 \times 10^7$ N C^{-1}, towards the 2.05 μC charge 1

2

Explanation: At the midpoint the two fields point in opposite directions.

(iii) the electric potential at the midpoint between the particles, [2]

Answer

$V = \dfrac{k(2.05 + 4.50) \times 10^{-6}}{0.0250}$ 1

$V = 2.36 \times 10^6$ V 1

2

Explanation: Potentials add as scalars.

(iv) the electric potential energy required to retain the particles in their respective position. [2]

Answer

$U = \dfrac{kQ_1Q_2}{r} = \dfrac{(9.0 \times 10^9)(2.05 \times 10^{-6})(4.50 \times 10^{-6})}{0.0500}$ 1

$U = 1.66$ J 1

2

Explanation: This is the work needed to bring the charges together from infinity.

(c) State Gauss's law. [1]

Answer

The total electric flux through any closed surface equals the net charge enclosed divided by ε_0: $\Phi = \frac{Q}{\varepsilon_0}$ 1

1

Explanation: The closed surface is a Gaussian surface.

(d) Two metal plates, each carries a uniformly distributed positive charge σ in isolation, are then placed parallel near to each other in the air. Determine the magnitude of the electric field

(i) outside the plates, [3]

Answer

Each plate alone gives a field $\frac{\sigma}{2\varepsilon_0}$ on both sides, directed away from it 1

Outside the plates the two fields are in the same direction 1

$E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$ 1

3

Explanation: Treat each plate as a large thin sheet of charge (Gauss's law with a pillbox).

(ii) between the plates. [2]

Answer

Between the plates the two fields are equal and opposite 1

$E = 0$ 1

2

Total for question **15**

Explanation: Like charges on both plates: no field between them.

19. (a)

(i) Define *capacitance* of a capacitor. [1]

Answer

The ratio of the charge on either plate to the p.d. between the plates, $C = \frac{Q}{V}$ 1

1

Explanation: Unit: farad.

(ii) Derive the capacitance for a parallel plate capacitor in terms of its physical properties. [3]

Answer

Field between the plates: $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$ 1

$V = Ed = \frac{Qd}{\varepsilon_0 A}$ 1

$C = \frac{Q}{V} = \frac{\varepsilon_0 A}{d}$ 1

3

Explanation: A is the plate area and d the separation.

(iii) Explain microscopically how a charged capacitor can increase its capacitance when a dielectric material is inserted between the plates. [3]

Answer

The molecules of the dielectric are polarised by the field (dipoles line up) 1

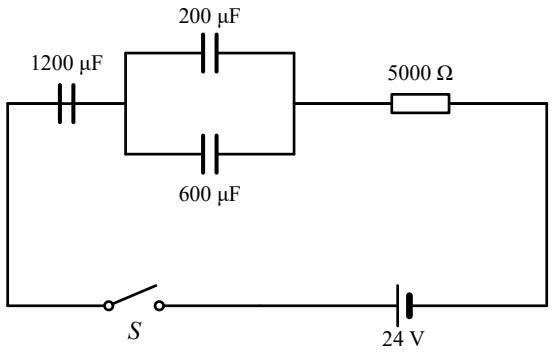
Induced surface charges on the dielectric produce a field opposing the applied field, so the field and the p.d. between the plates decrease 1

For the same charge, $C = \frac{Q}{V}$ increases 1

3

Explanation: C increases by the factor ε_r.

(b) A d.c. supply of 24 V which is connected to three capacitors, a resistor and a switch S is shown in the diagram below.



When switch S is closed, determine

(i) the time constant for the charging circuit, [4]

Answer

Parallel: $200 + 600 = 800\ \mu F$ 1

In series with 1200 μF: $\frac{(1200)(800)}{1200 + 800} = 480\ \mu F$ 1

$\tau = RC = (5000)(480 \times 10^{-6})$ 1

$\tau = 2.4$ s 1

4

Explanation: Find the total capacitance first.

(ii) the initial charging current, [2]

Answer

$I_0 = \frac{V}{R} = \frac{24}{5000}$ 1

$I_0 = 4.8 \times 10^{-3}$ A 1

2

Explanation: Initially the capacitors are uncharged, so all of the 24 V is across R.

(iii) the time taken for the charging current to drop to 1.6×10^{-3} A. [2]

Answer

$I = I_0 e^{-t/\tau} \Rightarrow t = \tau \ln \frac{I_0}{I} = 2.4 \ln \frac{4.8}{1.6}$ 1

$t = 2.6$ s 1

2

Total for question **15**

Explanation: The current falls to one third of its initial value.

20. (a) Explain the mechanism of electrical conduction in a conductor when it is connected to a battery. [3]

Answer

The battery sets up an electric field along the conductor 1

Free electrons, moving randomly, are accelerated opposite to the field between collisions with the lattice ions 1

They gain a small net drift velocity towards the positive terminal, producing a current 1

3

Explanation: Collisions transfer energy to the lattice (heating).

(b) A potential difference of 2.0 V is applied across a metal wire of length 24 cm and cross-sectional area 5.03×10^{-3} cm^2. A current of 1.5 A is produced.

(i) Determine the electric conductivity of the wire. [3]

Answer

$R = \frac{2.0}{1.5} = 1.33\ \Omega$; $A = 5.03 \times 10^{-7}$ m^2 1

$\sigma = \frac{L}{RA} = \frac{0.24}{(1.33)(5.03 \times 10^{-7})}$ 1

$\sigma = 3.58 \times 10^5\ \Omega^{-1}\ \text{m}^{-1}$ 1

3

Explanation: 1 cm^2 = 10^{-4} m^2.

(ii) The mean free time between collisions of free electrons with atoms is 1.78×10^{-15} s. Determine the number of free electrons per unit volume. [2]

Answer

$\sigma = \frac{ne^2\tau}{m} \Rightarrow n = \frac{\sigma m}{e^2\tau} = \frac{(3.58 \times 10^5)(9.11 \times 10^{-31})}{(1.60 \times 10^{-19})^2(1.78 \times 10^{-15})}$ 1

$n = 7.2 \times 10^{27}\ \text{m}^{-3}$ 1

2

Explanation: Uses the free-electron model of conduction.

(iii) Calculate the drift speed for the free electrons. [2]

Answer

$v_\text{d} = \frac{I}{nAe} = \frac{1.5}{(7.2 \times 10^{27})(5.03 \times 10^{-7})(1.60 \times 10^{-19})}$ 1

$v_\text{d} = 2.6 \times 10^{-3}\ \text{m s}^{-1}$ 1

2

Explanation: Equivalently $v_\text{d} = eE\tau/m$.

(iv) The typical speed of free electrons in a solid is in the order of 10^6 m s^{-1}. Explain the difference between typical speed and the drift speed in (b)(iii). [2]

Answer

The typical speed (about 10^6 m s^{-1}) is the speed of random thermal motion in all directions, with no net displacement	1
The drift speed is the small net (average) velocity along the field, superimposed on the random motion	1
	2

Explanation: Only the drift carries the current.

(c) Explain how the electric conductivity of a conductor depends on the change of temperature. [3]

Answer

As the temperature rises, the lattice ions vibrate with larger amplitude	1
Electrons collide more often, so the mean free time decreases (the number of free electrons in a metal stays almost constant)	1
$\sigma = \frac{ne^2\tau}{m}$ decreases: the conductivity falls	1
	3
Total for question	**15**

Explanation: In a semiconductor, n rises and the conductivity increases instead.